

FlowPilot: A Multi-Agent AI System for Continuous Flow Chemistry Process Design

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ABSTRACT

Translating a batch protocol into continuous flow requires chemical interpretation, quantitative engineering, and compatible equipment to be considered together. FlowPilot combines large language models (LLMs), standardized chemist intake, literature retrieval, deterministic calculation, and specialist council review to generate inventory-constrained process proposals. A shared design record links feed compositions, stage conditions, equipment assignments, and the process diagram. The workflow is supported by 464 classified literature records and 2,537 engineering rules. Across three chemistry cases, five generator models, and three repeats, matched one-shot and FlowPilot generation produced 90 outcomes evaluated using 14 fixed criteria and separate LLM judges. Mean benchmark scores were 0.794 for one-shot generation and 0.917 for FlowPilot; judge-reported critical flags totaled 98 and 4, respectively. Laboratory implementations of photoredox Giese addition/oxidation and DPDTC-mediated amidation gave 97–98% and 68–86% NMR yields, respectively, under the reported laboratory settings. These results demonstrate a traceable connection between reaction-level instructions, quantitative flow-process design, and laboratory implementation, while keeping computational design assessment distinct from measured chemical performance.

INTRODUCTION

Continuous-flow chemistry offers a versatile platform for synthesis by controlling how reagents are mixed, heated, irradiated, and brought into contact with gases or catalysts.¹⁻⁴ These capabilities have

expanded access to photochemical, electrochemical, and gas-liquid transformations and enabled practical routes to pharmaceutical intermediates.⁵⁻¹⁰ Connecting reaction stages can further reduce intermediate handling and support multistep or on-demand production.¹¹⁻¹⁴ Realizing these advantages, however, requires more than transferring a batch recipe into a smaller vessel. A successful translation must connect the chemistry to a suitable reactor arrangement, quantitative feed conditions, and the equipment available in the laboratory.¹⁵⁻¹⁶

Those choices are closely coupled. Reactor geometry influences heat and mass transfer, irradiation, and pressure loss, while an interstage addition changes both composition and the flow entering the next reactor.¹⁷⁻²¹ Gas-liquid processes introduce a further distinction between gas molar delivery and its pressure-dependent volume, making the definition of residence time particularly important.²² In a telescoped sequence, the composition leaving one stage also determines what the next stage can achieve. Integrated apixaban synthesis and photochemical routes to sartan intermediates illustrate the coordination of chemistry, hardware, and operating conditions required in multistep flow development.²³⁻²⁴

Digital methods increasingly support this development process. Computer-aided synthesis planning identifies reaction routes,²⁵⁻²⁷ process simulation evaluates connected unit operations,²⁸⁻²⁹ and Bayesian and active-learning methods guide experimental optimization.³⁰⁻³⁴ AI-informed robotic flow synthesis has linked planning to physical execution,³⁵ while RoboChem-Flex and UV-visible auto-reactometry extend modular optimization and LLM-assisted photochemical screening.³⁶⁻³⁷ These approaches demonstrate the value of integrating computation with experimentation. Preparing an initial configuration from a conventional reaction procedure remains an important entry point, especially when a laboratory must work within a specific set of pumps, reactors, and control modules.

Large language models provide an interface between such procedural descriptions and computational tools. ChemCrow and Coscientist combine chemical reasoning with tool use and experimentation, and El Agente coordinates computational chemistry workflows.³⁸⁻⁴⁰ Co-Scientist and Paper2Agent extend collaborative scientific reasoning and the reuse of computational methods.⁴¹⁻⁴² For laboratory procedures, ACRA links literature instructions to executable chemical descriptions and available hardware; AutoLabs combines clarification, calculation, and self-correction for liquid handling; and PRISM incorporates simulation feedback before robotic execution.⁴³⁻⁴⁵ Together, these studies show how language-based interaction can connect scientific intent to structured operations.

Several systems address flow chemistry directly. Chat-microreactor uses literature-derived information for flow-pattern prediction and microreactor design guidance, including a piperacillin application.⁴⁶

SapoMind combines condition recommendation with experimental optimization of continuous-flow lanolin saponification.⁴⁷ Ruan et al. developed LLM-RDF for end-to-end synthesis development, integrating experimental execution, kinetic analysis, optimization, purification, and reactor-design tasks.⁴⁸ These studies establish both quantitative engineering and experimental development as part of the emerging scope of chemical agents. For batch-to-flow translation, the practical task is to bring these decisions together in a connected process whose feed compositions, reactor conditions, and equipment assignments can be inspected as a whole.

Related work in chemical process design provides complementary approaches to this integration. Tian et al. generate process topologies and parameters for executable simulation flowsheets,⁴⁹ and the CeProAgents preprint organizes knowledge extraction, conceptual design, and parameter optimization into specialist agent groups.⁵⁰ The CAAF preprint investigates deterministic constraint checking and candidate revision, including a constructed flow-reactor design problem.⁵¹ The El Agente Gráfico and La Agente Óptima preprints from the Aspuru-Guzik group further explore structured execution and agent-supervised optimization.⁵²⁻⁵³ FlowPilot builds on this broader direction while focusing on the relationship between a reaction procedure and its laboratory-specific flow configuration: which streams enter each stage, how their compositions and flow rates determine operating conditions, and which available instruments can realize the connected design. **Table S1** provides the detailed literature comparison.

Assessing this capability requires more than fluent explanations or plausible individual settings. Chemical reasoning benchmarks have identified elementary errors and overconfidence in otherwise capable models,⁵⁴ and uncertainty-aware optimization studies have documented invalid suggestions within defined reaction search spaces.⁵⁵ For a flow process, the corresponding concern is whether the chemical sequence, material balance, reactor geometry, and operating procedure agree. Evaluation therefore needs to examine the consistency of the complete proposed process as well as its eventual chemical performance.

Here we present FlowPilot, a batch-to-flow design workflow that combines standardized chemist intake, chemistry-aware literature retrieval, deterministic engineering calculations, specialist council review, and inventory assignment. A shared final design record links the stream compositions and flow rates to stage conditions, equipment selections, and the process diagram. We evaluate retrieval alignment, compare delivered designs with matched one-shot generation using fixed criteria and separate LLM judges, and examine the effects of council components and candidate exploration. Two laboratory implementations then test complementary aspects of connected-process design: photoredox Giese addition followed by oxidation requires staged oxygen introduction, whereas DPDTC-mediated amidation requires interstage reagent addition and transfer of a reactive intermediate without isolation.

RESULTS

FlowPilot organizes batch-to-flow translation around a shared process description (**Figure 1**). The chemist supplies the protocol, design objective, and laboratory inventory, then resolves standardized follow-up questions about missing chemistry, constraints, and experimental history. The confirmed answers form a stored input package for the run. The upstream module converts that context into a chemistry plan identifying species, reaction stages, stream incompatibilities, sensitivities, and possible flow-specific advantages. Retrieval supplies relevant precedents, and engineering calculations establish an initial proposal containing reactor geometry, stream assignments, residence-time assumptions, and operating conditions. Council review and equipment reconciliation then assess that proposal before it becomes the final reported design.

The downstream council reviews alternatives across chemical and engineering domains (**Figure 1d**). The Designer proposes candidate operating points; Chemistry, Kinetics, Fluidics, and Safety specialists assess transformation fidelity, reaction-time plausibility, transport, equipment compatibility, and hazards. The Skeptic examines assumptions and contradictions, and the Orchestrator selects among the reviewed candidates. The retained record includes revisions, rejected alternatives, and selection rationales, allowing the final proposal to be inspected alongside the decisions that produced it.

The selected proposal is reconciled into a common final design record containing stage and stream parameters, equipment assignments, and the process graph. Numerical views and the diagram are generated from this record, while unresolved requirements and earlier calculations remain available as diagnostics. This arrangement links proposed conditions to specific equipment and preserves the context for later experimental feedback. **Figures S1–S2** and **Tables S2–S3** describe the data contracts and provenance mechanisms.

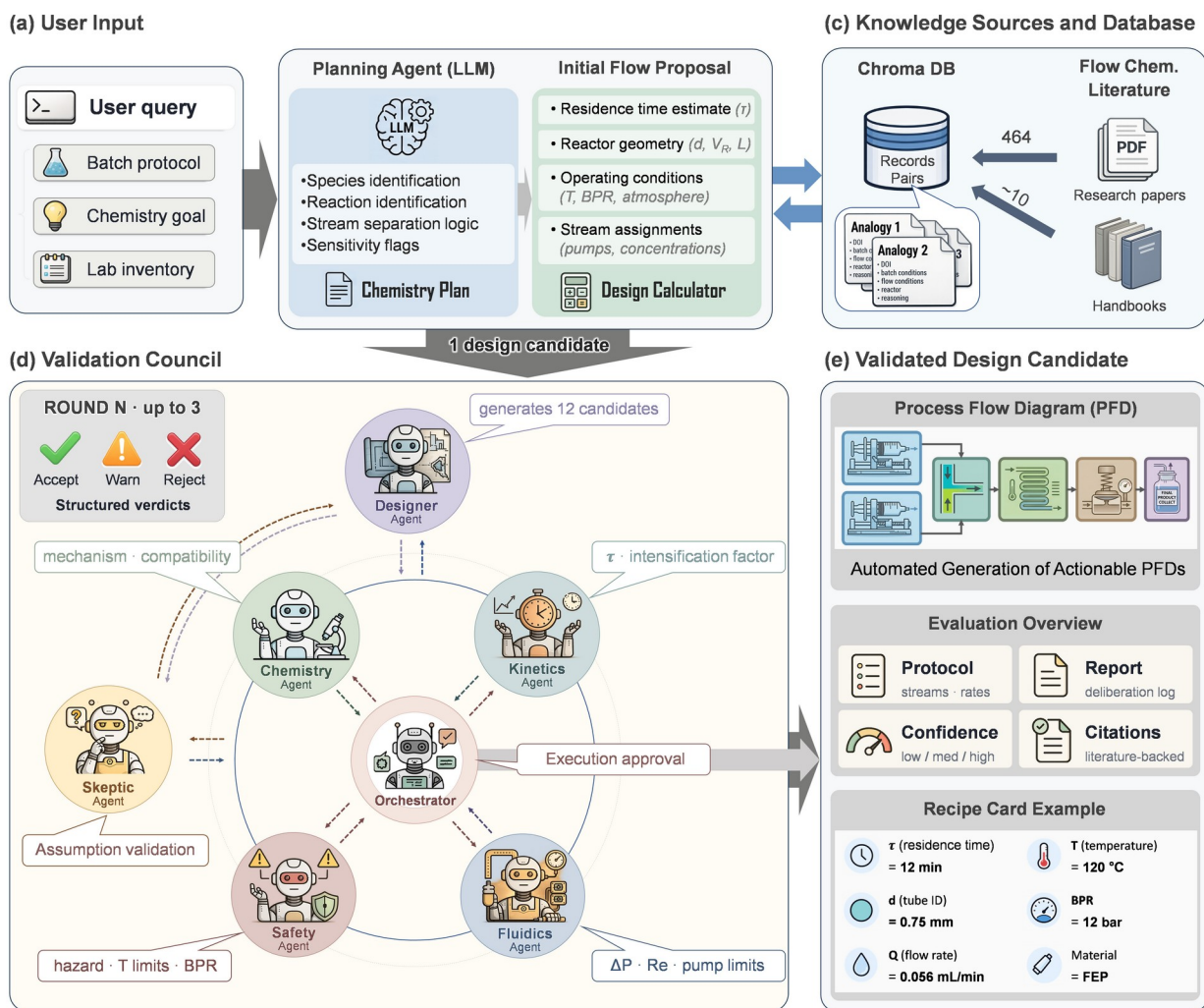


Figure 1. FlowPilot system architecture. User inputs are converted into a chemistry plan and an initial engineering proposal, supported by flow-chemistry literature and handbook knowledge. The seven-agent council generates and reviews candidate designs, records structured verdicts, and proposes bounded revisions. The selected candidate is reconciled with engineering calculations and laboratory inventory before a process-flow diagram, protocol, report, confidence assessment, and supporting citations are returned. The illustrated recipe card is schematic; computational validation does not establish experimental performance.

Literature and rule-based coverage

The literature dataset covers a broad range of flow-chemistry applications (**Figure 2**). The curated export contains 464 classified literature records spanning thermal synthesis, photoredox catalysis, heterogeneous photocatalysis, cross-coupling, oxidation/reduction, and other reaction classes. Reactor types range from

capillary/coil and microreactor/chip systems to photoreactors and packed beds, with fluoropolymer tubing prominent among the reported materials. Explicit inlet-stream counts are available for 154 processes; two- and three-stream configurations account for 91 and 30 of these records, respectively. These annotations supply precedents rather than prescribing a default number of streams. The diversity supports retrieval across photochemical and non-photochemical processes with different materials and operating regimes.

The engineering knowledge base contained 2,537 structured rules spanning reactor design, heat transfer, catalyst behavior, residence time, photochemistry, materials, scale-up, pressure, mixing, and safety (**Figure 2e-f**). Coverage was not uniform across chemistry classes: heat-transfer rules were densely associated with thermal synthesis and cross-coupling, whereas photochemistry rules concentrated in photoredox and photocycloaddition classes. This structured rule landscape supplies domain context for evaluating model-generated designs alongside deterministic checks, rather than relying only on model priors. The frozen corpus composition and computed rule-base coverage are reported in **Figures S3–S4** and **Tables S4–S5**.



Figure 2. Literature corpus and engineering rule-base coverage. (a-d) Reaction classes, reactor types, reported materials, and inlet-stream configurations in the curated flow-chemistry corpus; explicit inlet-stream counts are available for 154 records. (e) Category and severity distribution within the 2,537-rule engineering knowledge base. (f) Recorded associations between rule categories and chemistry classes. Full categorical exports and field-specific availability are reported in **Figures S3–S4** and **Tables S4–S5**; associations indicate knowledge coverage rather than independent validation of every rule.

Plan-aware retrieval and metadata alignment

FlowPilot uses the chemistry plan to enrich retrieval queries with reaction class, mechanism, catalyst family, solvent, and wavelength information (**Figure 3a**). The operational search progresses from mechanism- and phase-filtered paired records to relaxed filters and, when necessary, the full corpus; a

field-aware score ranks the candidates (**Figure S5**; **Table S6**). We examined the contribution of metadata in separate retrospective analyses using stored corpus embeddings. These analyses isolate query enrichment and reranking behavior rather than testing the complete live retrieval workflow.

In the two query-enrichment examples, chemically specific term counts increased from 7 to 15 and from 7 to 14 (**Figure 3b**). A separate rank analysis evaluated 20 neighbors for each of 80 corpus queries, giving 1,600 query–result pairs. Adding the metadata score changed 401 ranks (25.1%) relative to embedding-only ordering. These changes quantify reranking activity, not an independent measure of reaction relevance.

In the family-alignment analysis, the mean proportion of top-five records sharing the query’s recorded catalyst family increased from 40.0% to 95.0% for iridium, 42.7% to 90.7% for organic dyes, 77.3% to 94.7% for ruthenium, 87.1% to 98.6% for TiO₂, and 45.0% to 60.0% for ZnO (**Figure 3c**). The corresponding query counts were 8, 15, 15, 14, and 4. Catalyst-family metadata also contribute to the reranking score, so this is a descriptive alignment test, not independent validation of mechanistic transfer or predicted yield. **Figure 3d** illustrates the resulting ranking change; **Figures S5–S6** and **Tables S6–S7** provide the workflow, analysis settings, and applicability limits.

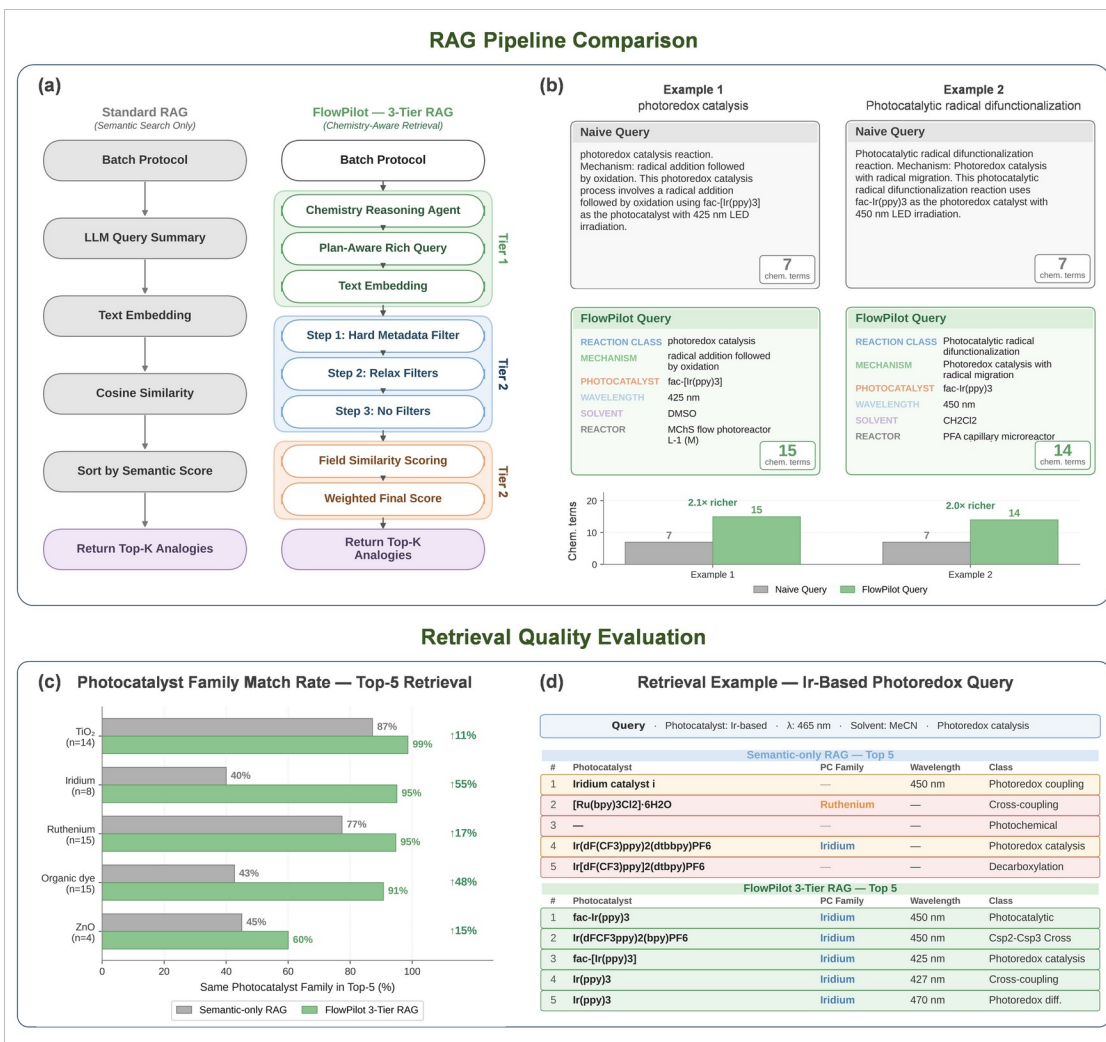


Figure 3. Retrieval workflow and retrospective metadata-alignment analyses. (a) Operational query enrichment, metadata-filter relaxation, and field-aware ranking. (b) Two examples of query enrichment, showing counts of chemically specific terms. (c) Mean fraction of the top five retrieved records with the same recorded photocatalyst family as the query; *n* denotes query count. (d) An illustrative iridium-query ranking. The analyses in (c–d) use stored embeddings and the offline scoring implementation described in Methods and ESI Section 3.4. Missing family labels are unassigned metadata, not verified chemical mismatches; displayed percentage gains are percentage-point differences.

Ablation and architecture comparison

To evaluate the delivered designs, we compared one-shot generation with the full FlowPilot pipeline under the same chemistry, inventory, and requested output contract (**Figure 4a**). The cases covered copper(I)-catalyzed azide–alkyne cycloaddition (CuAAC), photochemical oxidation, and hydrogenolysis,

with three generation repeats per model–architecture–case combination. Separate Qwen, OpenAI, and Claude evaluator calls scored normalized final designs against 14 fixed criteria covering chemical fidelity, material and reactor consistency, inventory, operating procedures, and uncertainty. Applicable integer ratings from 0 to 4 were averaged with equal criterion and judge weights and normalized to a score from 0 to 1; critical flags were recorded separately. **Tables S8–S10** define the rubric and module screen.

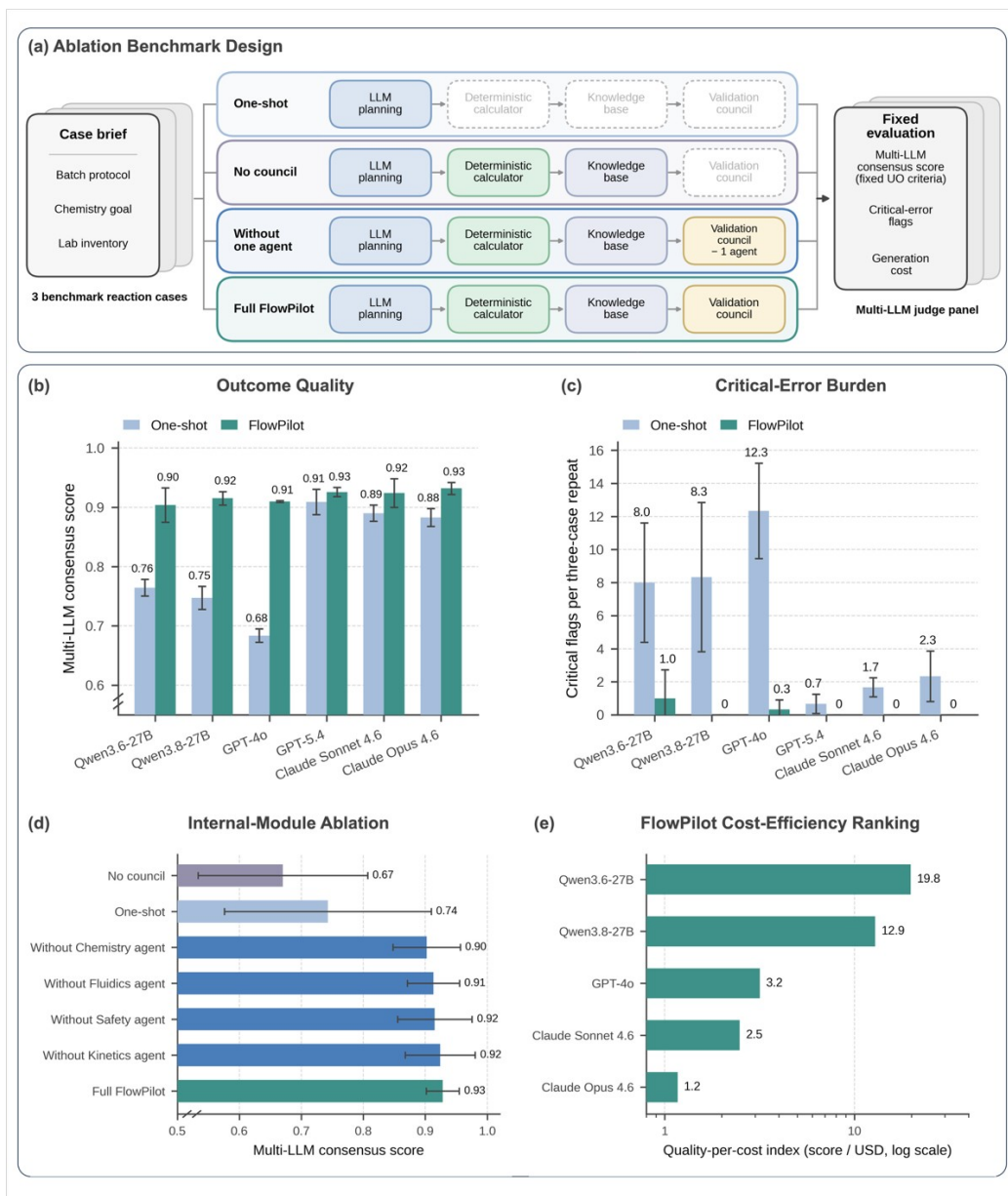


Figure 4. Ablation benchmarking and cost efficiency. (a) Architecture comparison under shared inputs and fixed evaluation. (b) Benchmark scores and (c) raw judge critical flags: mean and sample SD across three repeats, each containing three chemistry cases. The six-model display includes the archived GPT-5.4 generator comparison; this additional model is not included in the five-model aggregates in **Tables S11–S12**. (d) Selected Qwen3.8-27B ablations: mean and between-case SD with one generation per case-condition (all 15 conditions: **Table S10**). (e) Mean FlowPilot score divided by mean generation cost under the recorded price schedule, excluding judging and deployment. Critical flags are judge findings, not independently confirmed physical errors. ESI Section 4 provides evaluation details.

Across the five-model cohort in **Table S11**, full-pipeline mean scores ranged from 0.904 to 0.932, compared with 0.684 to 0.890 for matched one-shot generation (**Figure 4b**). The mean gain was largest for GPT-4o (+0.226), followed by Qwen3.8-27B (+0.168) and Qwen3.6-27B (+0.139); the Claude gains

were smaller (+0.034 and +0.049). The judges recorded 98 critical flags across the 45 one-shot outcomes and 4 across the 45 FlowPilot outcomes; 18 and 43 outcomes, respectively, had no critical flag (**Figure 4c**; **Table S12**). The score and flag summaries therefore point in the same direction within this cohort: improved completeness and fewer flagged inconsistencies. They do not establish universal model superiority or measured reaction success. **Figure 4d** provides a complementary internal comparison, with council removal causing the largest score decrease among the configurations shown.

Chemistry-specific scores are shown in **Figure S7**, criterion-level differences in **Figure S8**, and run-level critical flags in **Figure S9**. **Figure S10** reports generation tokens, estimated costs and runtime. The score-per-cost values use the archived token-price schedule, not total deployment costs. **Table S13** defines the numerical closure checks applied to stage and feed calculations.

A separate council model-matrix and candidate-budget study used an integrated isoxazole synthesis case⁵⁶ (**Figure S11**; **Tables S14–S17**). Across five repeats, the normalized engineering radar-area score changed from 0.31 ± 0.03 before council review to 0.48 ± 0.23 afterward. This geometric summary of engineering indicators is distinct from the fixed-criteria LLM-judge score. The archived model matrix, revision counts, and normalization definitions document configuration sensitivity.

The interface links these assessments to the chemist's workflow. **Figures S12–S13** show protocol entry and fixed-ID follow-up questions. **Figure S14** and **Table S18** document inventory review, while **Figure S15** and **Table S19** show stage-resolved output. **Table S20** retains concrete one-shot and FlowPilot problems, including an evaluator false positive; **Figure S16** and **Table S21** show council disagreement and selection. These records make assumptions and unresolved requirements inspectable before laboratory implementation.

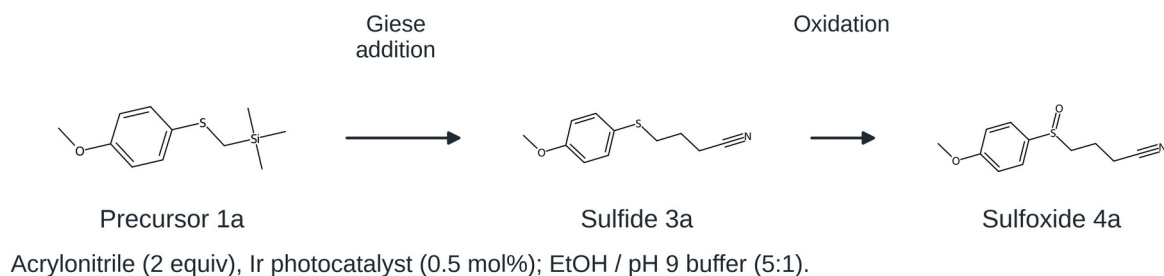
Case study 1: photoredox Giese addition followed by aerobic oxidation

The first laboratory case couples photoredox Giese addition with oxidation of the resulting sulfide to a sulfoxide (**Figure 5**). The reported chemistry uses photoinduced electron transfer to generate an alpha-thiomethyl radical from an alpha-silyl sulfide, followed by addition to acrylonitrile and a separate oxygen-dependent oxidation step.⁵⁷ Its process-design requirement is the transition from an oxygen-free first stage to an oxidizing second stage within one connected apparatus. The batch input specified 4 h under argon followed by 6 h in air. The NMR-yield comparison uses the published flow result, providing context for implementation rather than a controlled test of identical hardware.

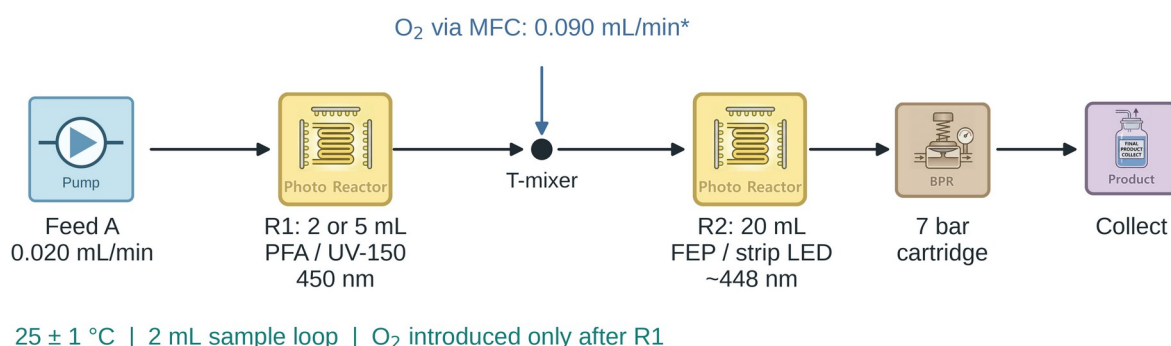
Kyung Hee University (KHU) implemented the proposal using a 2 or 5 mL PFA coil in a Vapourtec UV-150 module followed by a separately illuminated 20 mL FEP coil, both at 25 ± 1 °C. A 2.0 mL sample loop introduced substrate 1a (0.100 M), acrylonitrile (0.200 M), and iridium photocatalyst (0.500 mM) into a carrier stream at 0.020 mL min⁻¹. The nominal first-stage liquid residence times were therefore 100 and 250 min. Oxygen entered at the interstage mixer at a controller setting of 0.090 mL min⁻¹. Under the proposal's STP convention (273.15 K, 1 atm), this setting corresponds to 0.00402 mmol min⁻¹ of oxygen, or 2.01 equivalents relative to the nominal substrate feed. The resulting second-stage inlet-reference index, $20/(0.020 + 0.090)$, is 181.82 min; it is not a physical residence time at operating pressure. The laboratory gas-reference conditions and sample-plug collection details are distinguished from these nominal calculations in ESI Section 9.

KHU initially observed backflow toward the first reactor with pure oxygen at 0.43 mL min⁻¹ and liquid at 0.020 mL min⁻¹. The revised point retained pure oxygen at 0.090 mL min⁻¹. A gas-line check valve was included in the proposal, but its presence alone does not establish protection of the upstream liquid branch. The installed BPR was a 7 bar cartridge; recorded system pressures were 5.1–5.6 bar. Sets 1/3, reported together for the 2 mL first reactor, gave 97% yield; Set 2, with a 5 mL first reactor, gave 98%. These and the 90% literature flow reference are NMR yields.⁵⁷ The difference between KHU configurations is too small to attribute to residence time without replicates, and the comparison with literature is not matched for operating conditions. **Table S22** records the KHU settings; **Figures S17–S18** show irradiation sources and the assembled apparatus.

a Two-stage photoredox sequence



b Connected configuration implemented at KHU



c Operating points and yield

KHU set(s)	R1 time min	R2 time* min	Pressure bar	Yield (%)
1 / 3	100	181.8	5.1–5.4	Literature flow 90
				sets 1 / 3 97
2	250	181.8	5.2–5.6	set 2 98

*R2 is an inlet/STP time index, not operating-pressure residence time.

Figure 5. Photoredox Giese addition/oxidation from chemistry to laboratory implementation. (a) Precursor 1a is converted through sulfide 3a to sulfoxide 4a. (b) Connected equipment arrangement; oxygen is introduced after R1. Feed A contains 0.100 M 1a, 0.200 M acrylonitrile and 0.500 mM iridium photocatalyst. The process schematic is not a piping or safety-certification diagram. (c) KHU operating points and yield, with the literature flow reference from Park et al.⁵⁷ All plotted yields are determined by ¹H NMR; grouped Sets 1/3 are one supplied result entry, not independent repeats. R1 time is V/Q_{liquid}. The R2 value is the nominal inlet/STP index V/(Q_{liquid} + Q_{gas,STP}), not residence time under operating pressure. The MFC reference state requires confirmation. Literature and KHU experiments used different conditions.

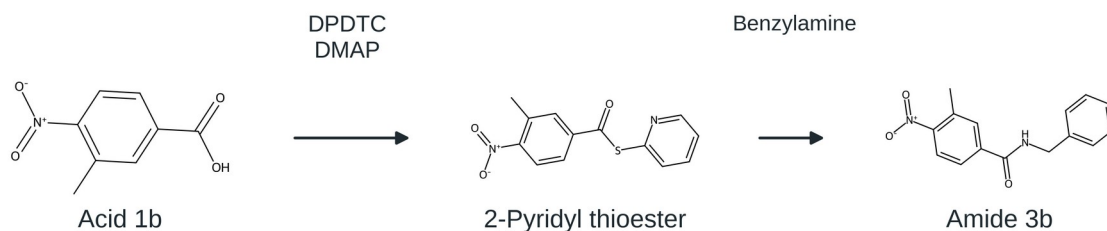
Case study 2: telescoped DPDTC-mediated amidation

The second laboratory case is the thermal amidation of 3-methyl-4-nitrobenzoic acid with benzylamine (**Figure 6**). Activation with 2,2'-dipyridyldithiocarbonate (DPDTC) in the presence of DMAP generates a 2-pyridyl thioester, which undergoes downstream aminolysis to form the amide.⁵⁸ The batch input specifies two 30 min stages at 95 °C, separated by cooling and amine addition. In the connected process, the intermediate is transferred without isolation, so its formation and downstream consumption must be considered together when choosing feed ratios and stage times.

KHU implemented the proposed separate-feed arrangement using two channels of the Vapourtec E-series system. Feed A contained acid (0.500 M), DPDTC (0.525 M), and DMAP (0.0500 M) in 2-MeTHF; Feed B contained benzylamine (2.10 M). In Set 1, $Q_A = 0.160$ and $Q_B = 0.0400$ mL min⁻¹ correspond to nominal acid and amine molar flows of 0.0800 and 0.0840 mmol min⁻¹, preserving 1.05 equivalents of amine. Two 5 mL ETFE coils give nominal stage times of 31.25 and 25.00 min because the second reactor receives both feeds. The reported implementation used a 2.0 mL sample loop for Feed A, direct interstage addition, and a 95 °C water bath for both coils. A 7 bar cartridge BPR was fitted; the recorded pressure for Set 1 was 5.2–5.3 bar. Thus hardware selection, nominal stage times, and measured pressure remain distinct parts of the process record.

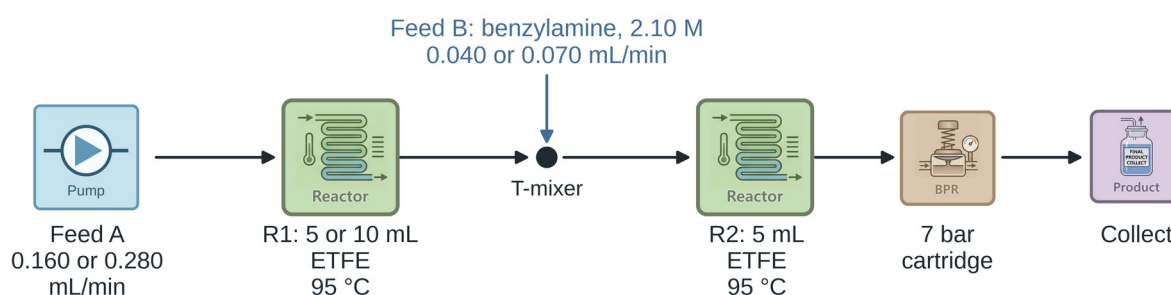
Set 1 gave 86% yield. Sets 2/3 were reported together for a 10 mL first reactor and 5 mL second reactor at feed rates of 0.280 and 0.0700 mL min⁻¹, giving stage times of 35.71 and 14.29 min and a yield of 68%. Both outcomes are NMR yields, as is the 98% literature batch reference plotted in **Figure 6**.⁵⁸ Relative to Set 1, the total nominal stage time decreased from 56.25 to 50.00 min while the nominal substrate feed increased from 0.0800 to 0.140 mmol min⁻¹. The lower yield at higher flow reinforces the need to evaluate the connected sequence, but does not isolate a kinetic cause because several operating variables changed together. **Table S23** and **Figure S19** report the operating points and apparatus; **Tables S24–S25** document implementation changes and component molar feeds.

a Telescoped activation and amidation



DPDTC (1.05 equiv), DMAP (10 mol%), benzylamine (1.05 equiv); 2-MeTHF.

b Connected configuration implemented at KHU



Two E-series pump channels | 2 mL sample loop | No intermediate isolation

c Operating points and yield

KHU set(s)	R1 time min	R2 time min	Pressure bar	Yield (%)
				Literature batch 98
1	31.25	25.00	5.2–5.3	set 1 86
2 / 3	35.71	14.29	5.5–5.8	sets 2 / 3 68

Stage times use V_1/Q_A and $V_2/(Q_A + Q_B)$, respectively.

Figure 6. DPDTC-mediated amidation from reaction sequence to connected operation. (a) Acid 1b is activated to a 2-pyridyl thioester and undergoes aminolysis to amide 3b. (b) Two heated ETFE coils receive acid/DPDTC/DMAP in Feed A (0.500/0.525/0.0500 M) and benzylamine in Feed B (2.10 M), added between the stages. (c) KHU operating points and yield, with the literature batch reference from Saunders et al.⁵⁸ All plotted yields are determined by ¹H NMR. Sets 2/3 share one supplied result entry. Stage times use V_1/Q_A and $V_2/(Q_A + Q_B)$; recorded pressures are distinct from the 7 bar BPR cartridge rating. The literature reference is contextual rather than a matched flow-control experiment.

DISCUSSION

FlowPilot treats batch-to-flow translation as a connected-process design problem. Its principal contribution is to link reaction interpretation to feed compositions, stage calculations, and inventory assignments in a common record. This makes an operating proposal inspectable across chemistry and engineering domains, rather than relying on agreement between separately generated numerical tables and diagrams. The task complements microreactor assistants, synthesis-development platforms, and process-simulation agents, while providing a structured starting point for subsequent optimization.^{46-51,59-60}

The architecture comparison supports this approach within the tested cases: FlowPilot had higher mean scores than matched one-shot generation for each of the five retained generators, although the size of the difference varied by model (**Figure 4b**; **Table S11**). In the Qwen3.8-27B module screen, removing the council reduced the mean score from 0.928 to 0.670, whereas individual specialist removals had smaller effects (**Figure 4d**; **Table S10**). The single generation per case-condition supports descriptive module attribution, not a statistical ranking of every component. The separate candidate-budget study likewise shows that wider exploration does not automatically improve the selected design (**Figure S11**; **Tables S14–S17**).

The laboratory studies examine aspects not resolved by a design score. The Giese/oxidation sequence requires spatial separation of the oxygen-free and oxygen-fed stages; the backflow observation also shows that gas equivalents and volume-flow arithmetic alone are insufficient to establish operability. The amidation sequence links thioester formation, interstage amine addition, and downstream conversion without isolation. Its different yields at the two operating points demonstrate the importance of evaluating the connected process rather than either reactor in isolation. These sample-loop experiments establish product formation under the reported settings, while sustained reagent-fed operation and experimental repeatability require separate evaluation.

Retaining design conflicts and experimental deviations is therefore an important part of the workflow. The one-shot and FlowPilot inconsistencies documented in **Table S20** identify where further checking is needed; the record links such findings to the inputs, calculations, and review decisions that produced them. Measured outcomes can subsequently be returned with the actual operating conditions to guide revision. Combining this record with Bayesian optimization or language-guided experimental priors offers a route to prospective closed-loop development,^{30-34,61} rather than treating a numerically consistent first proposal as an optimized reaction.

CONCLUSION

FlowPilot addresses batch-to-flow translation by treating chemistry, stream composition, reactor operation and available equipment as parts of one connected design. Standardized intake makes the chemist's objectives and constraints explicit; retrieval and specialist review supply chemical context; deterministic calculations reconcile the quantities that must agree. A shared final record links the resulting stream and stage conditions to equipment assignments and the process diagram, allowing the proposal to be inspected and revised rather than accepted solely as a fluent explanation.

The computational and laboratory evaluations test complementary aspects of this approach. Across the retained five-model, three-chemistry comparison, FlowPilot achieved a mean fixed-criteria score of 0.917, compared with 0.794 for matched one-shot generation, and received fewer judge-reported critical flags. The module screen identifies the complete council as an important contributor within those cases, without establishing that every specialist or larger candidate budget always improves the result. The two laboratory implementations extend the assessment to connected chemistry: staged oxygen delivery supported Giese addition/oxidation at 97–98% yield, while activation followed by amidation gave 68–86% yield. These values were determined by NMR and represent the reported operating points, not optimized or replicated performance across a broad reaction space.

The practical contribution is therefore a traceable starting point for experimental process development, not a replacement for it. The observed backflow and the yield change between amidation conditions show why apparatus behavior and complete-sequence performance must remain part of evaluation. Returning actual operating conditions and measured outcomes to the same design record provides a basis for subsequent refinement. Further prospective studies can test whether this design-and-feedback workflow reduces experimental effort across a broader range of chemistries and laboratory inventories.

METHODS

Standardized inputs and study organization

The standardized workflow begins with a batch protocol and a structured intake package recording the objective, chemistry identity, experimental history, laboratory inventory, operating limits, and chemist hypotheses. Follow-up questions use stable identifiers and predefined readiness rules; optional evidence can be explicitly marked unavailable. The confirmed package is frozen for the run, retaining the answers and their provenance. Measured observations, protocol facts, and hypotheses are stored separately. Hard safety and equipment constraints remain mandatory even when earlier measurements were obtained under

different conditions. Computational comparisons used archived case packages rather than newly generated conversations, keeping the supplied chemistry and constraints constant between architectures. The intake contract and question rules are given in ESI Section 2 and **Tables S2–S3**. Benchmark campaigns and subsequent laboratory-design runs were treated as distinct software/configuration records; the dated development history is summarized in the companion version-history record.

Literature corpus and plan-aware retrieval

The literature resource comprised 464 classified flow-chemistry records annotated with available reaction, reactor, material, operating-condition, yield, and inlet-stream information. Missing metadata were retained as missing rather than filled from model inference. A complementary handbook-derived store contained 2,537 engineering rules with topic, severity, source, and applicability information.^{3–4,21} Relevant rules supplied context to chemistry interpretation; only checks implemented in the calculation and validation routines acted as deterministic constraints. Field completeness and rule coverage are reported in **Tables S4–S5**.

For operational retrieval, the chemistry plan supplies reaction class, mechanism, catalyst, solvent, phase regime, and operating conditions. The search begins with paired batch-flow records under mechanism and phase filters, relaxes filters when fewer than three candidates remain, and expands to the full corpus when no paired result remains. The final score combines semantic and field similarity with weights of 0.60 and 0.40. The operational field components and weights are listed in **Table S6**. In the architecture benchmark, the designated source record was excluded before final retrieval selection; this restriction does not establish absence from model pretraining.

The retrospective figure analyses used stored corpus embeddings and inverse Euclidean-distance similarity, $1/(1 + d)$, rather than newly generated query embeddings. The rank analysis reranked the 20 nearest non-self records for 80 queries using photocatalyst, solvent, and wavelength terms weighted 0.30, 0.20, and 0.20; temperature and concentration were not included in that offline score. A separate family analysis sampled up to 15 queries per recorded family and measured the fraction of matching-family records in each top-five result set. Its 56 queries and script-specific similarity functions are documented in ESI Section 3.4 and **Table S7**. These analyses measure metadata alignment, not the full live retrieval policy or independent chemical correctness.

Engineering calculations and stage accounting

The engineering layer recalculated quantities dependent on proposed operating choices. Component molar delivery was obtained from concentration and liquid volumetric flow, and each stage received the

cumulative flow of the streams introduced at or upstream of that stage. For a nominal liquid-filled reactor, the stage time was calculated as reactor volume divided by local liquid flow. An interstage feed therefore changed the downstream material balance and time calculation without changing the upstream flow. Packed-bed empty volume, estimated liquid holdup, and catalyst-contact time were distinguished where applicable. Stage totals were formed only from times reported on compatible bases.

Gas equivalents were calculated from the reactive-gas molar flow divided by the limiting-reactant molar flow, including gas composition and the stated reference temperature and pressure. The proposal STP convention was 273.15 K and 1 atm. Where required by a benchmark contract, in-channel gas volume was derived by ideal-gas scaling using absolute operating pressure and temperature. The inlet-reference index $V/(Q_{\text{liquid}} + Q_{\text{gas,STP}})$ was kept separate from pressure-corrected apparent time and from an experimentally measured residence time. Pressure-loss and thermal screening used the selected geometry, phase assumptions, and available material properties; estimated heat-transfer or reaction-enthalpy inputs were identified as assumptions. The implemented quantities, multiphase conventions, and heat-transfer screening are detailed in ESI Section 5 and **Table S13**.

Candidate generation and council review

The downstream Designer generated alternative geometries and operating choices from the chemistry plan, retrieved analogies, engineering calculations, and inventory constraints. The Chemistry, Kinetics, Fluidics, and Safety specialists reviewed chemical compatibility, reaction-time plausibility, transport and hardware feasibility, and hazards, respectively. The Skeptic examined assumptions and contradictions, and the Orchestrator selected among the reviewed candidates. Agent verdicts and proposed revisions were stored with candidate identifiers. Changes to independent variables triggered deterministic recalculation before a candidate could be compared or published. Candidate and revision budgets followed the configuration recorded for each campaign rather than being inferred from the final output. Rejected alternatives and revision history were retained alongside the selected candidate. Council terminology, configuration studies, and examples of actual deliberation are documented in ESI Section 6 and Section 7.5.

Inventory realization and result publication

Inventory profiles were imported as structured data or normalized from laboratory documents into typed equipment records with stable identifiers. Available quantities, system membership, material, reactor volume and diameter, pump range, pressure and temperature limits, and compatible light modules were recorded where supplied. Realization assigned required operations to available items and checked the

declared capability and connectivity constraints. Calculations were repeated when equipment selection changed a dependent quantity. An unresolved essential requirement was retained as a diagnostic requirement, not silently replaced with invented hardware (**Table S18**).

The final design contract reconciled the selected proposal, stage table, stream table, equipment allocation, and topology. GUI numerical views and the executable process diagram were derived from that record; earlier calculations and council exchanges remained separately available for inspection. Timestamped run folders stored the input package, result, final contract, topology, equipment allocation, and available diagram artifacts. Experimental feedback was linked to the conditions actually tested before being supplied to a later design cycle. ESI Sections 2.4–2.5 and 7 document the GUI and saved artifacts, with the machine-readable feedback and provenance records retained in the companion archive. Exact computational replay additionally requires the model, decoding settings, dependencies, and software revision recorded for the campaign.

Matched architecture comparison and LLM judging

The primary comparison used CuAAC, photochemical oxidation, and hydrogenolysis case packages with the same protocol, objective, inventory, and final-output contract for both architectures. One-shot generation received this information in a single call; FlowPilot distributed it across its interpretation, calculation, review, and realization stages. Qwen3.6-27B, Qwen3.8-27B, GPT-4o, Claude Sonnet 4.6, and Claude Opus 4.6 were each used with both architectures for three generation repeats per case, giving 90 outcomes at the recorded generation temperature of 0.2. The archived GPT-5.4 generator results additionally displayed in **Figure 4b–c** were not included in these five-model aggregates or **Tables S11–S12**. Exact case prompts, inventories, and model records are retained in ESI Section 4. The comparison concerns delivered design quality under matched task information, not equal token or computation budgets.

For judging, final designs were normalized into candidate packets with generator and architecture labels removed. Separate Qwen, OpenAI, and Claude judge calls received the candidate, common input context, reference facts, deterministic verification evidence, and the fixed 14-criterion rubric. The archived judge configuration used Qwen3.6-27B, GPT-5.4, and Claude Sonnet 4.6 at temperature 0.0; these evaluator calls were separate from candidate generation. Each applicable criterion received an integer rating from 0 to 4. Ratings were averaged across judges for each criterion, then across applicable criteria, and divided by four to give a score from 0 to 1. All judges and applicable criteria had equal weight. Gas bookkeeping and multistage closure could be excluded only when their applicability conditions were false; missing required information was scored rather than treated as inapplicable. Critical flags were counted separately

from the numerical score. Deterministic evidence did not confer an architecture-specific bonus, and identity masking did not guarantee that output style concealed the architecture. Some evaluator model families also served as generators; separate calls therefore do not establish statistical independence or remove possible model-family preferences. Scoring anchors, criterion definitions, and applicability rules are specified in **Tables S8–S9**.

Ablation experiments and statistical reporting

For each generator and architecture, the three case scores were averaged within each repeat. Reported architecture error bars are the sample standard deviation of these three repeat-level means, not the dispersion of all nine individual outcomes. Critical-flag counts were summed across the three cases within a repeat before calculating their mean and sample standard deviation. The aggregate values in **Tables S11–S12** use only the five-model cohort. Generation cost was estimated from recorded token usage and the archived model-specific price schedule, excluding evaluator calls and deployment costs. **Figure 4e** uses mean score divided by mean generation cost; the case-specific ESI plots instead summarize per-outcome score-to-cost ratios (**Figure S10**). The comparisons are descriptive; no statistical-significance claim or population-wide generalization is made from three repeats and three selected chemistries.

The internal ablation screen fixed the generator to Qwen3.8-27B and tested 15 conditions over the same three cases, producing 45 outcomes with one generation per condition-case pair. The conditions removed or modified council components, revision stages, selection, or candidate budgets. Disabled specialists received no substitute score, and active council weights were renormalized. The resulting standard deviations describe between-case variation, not repeatability; all conditions are reported in **Table S10**. A separate historical study crossed four upstream and four council endpoints and tested candidate budgets of 1, 6, 12, and 24 with five repeats. Its engineering-profile metrics were normalized independently of the LLM-judge rubric (ESI Section 6; **Tables S14–S17**). No experimental yield uncertainty was inferred from these computational repeats.

Laboratory implementation and chemical analysis

KHU implemented the two connected-stage cases using a Vapourtec E-series system with sample-loop injection. For the Giese addition/oxidation sequence, a PFA coil in a 450 nm UV-150 module was followed by a separately illuminated FEP coil. The first stage was designed to remain oxygen-free, although the exact feed-degassing procedure was not specified in the supplied laboratory methods. Pure oxygen was introduced through an MFC at the interstage mixer, and both reactors were maintained at 25 ± 1 °C. For DPDTC-mediated amidation, the acid/DPDTC/DMAP solution and benzylamine solution

were delivered by separate channels, with amine addition between two ETFE coils maintained at 95 °C. Both setups used a 7 bar cartridge BPR; recorded system pressure was reported separately from the cartridge rating. Actual concentrations, molar feeds, operating points, and proposal-to-implementation differences are tabulated in ESI Section 9 and **Tables S22–S25**.

NMR yields were reported using 1,3-benzodioxole as the internal standard. Isolated products were characterized by ^1H and ^{13}C NMR spectroscopy and low-resolution mass spectrometry. Shared response-set entries in the supplied laboratory record were treated as grouped operating points, not independent replicates, and no experimental standard deviation was assigned without individual measurements. ESI Section 9 contains the feed preparation, work-up, irradiation specifications, apparatus photographs, and characterization records; **Figures S20–S23** reproduce the product spectra. MFC reference-state confirmation, collection windows, and quantitative-NMR calculation records are identified there as outstanding metadata rather than filled by inference.

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A.M. and M.P. contributed equally to this work. The complete contributor-role statement and approval of the final author list will be confirmed by all authors before submission.

Availability of data and code

Source code is publicly accessible at <https://github.com/arntfgh/flowpilot>. Benchmark scripts and the data-access index are available at https://github.com/arntfgh/flowpilot/tree/master/ablation_test and https://github.com/arntfgh/flowpilot/tree/master/ablation_results, respectively. The latter is currently an index, not the complete outcome archive. The ESI contains the reported numerical summaries, experimental procedures, and characterization spectra. As of 29 September 2026, the repository is public but has no tagged manuscript release; the complete benchmark archive, raw experimental analytical files, and a persistent data DOI have not yet been deposited. These release items remain to be completed before submission.

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ABBREVIATIONS

BPR, back-pressure regulator; CuAAC, copper(I)-catalyzed azide–alkyne cycloaddition; DPDTTC, 2,2'-dipyridyldithiocarbonate; DMAP, 4-(dimethylamino)pyridine; Da, Damköhler number; ETFE, ethylene tetrafluoroethylene; FEP, fluorinated ethylene propylene; IF, intensification factor; KHU, Kyung Hee University; LLM, large language model; LRMS, low-resolution mass spectrometry; MFC, mass flow controller; 2-MeTHF, 2-methyltetrahydrofuran; NMR, nuclear magnetic resonance; Pe, Péclet number; PFA, perfluoroalkoxy; PFD, process flow diagram; PTFE, polytetrafluoroethylene; RAG, retrieval-augmented generation; Re, Reynolds number; STP, standard temperature and pressure (273.15 K and 1 atm for the proposal calculations); STY, space-time yield.

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